FLIGHT, ANIMAL

THE ONLY ANIMALS CAPABLE of powered flight are birds, bats, and insects. Some other animals can glide for short distances. Flight is very useful. It helps the animals to find food, escape from predators, and migrate long distances. Flying animals need wings, powerful wing muscles, a streamlined shape, and a lightweight body. They also need to eat lots of food to give them the energy to flap their wings.

Birds

A bird's wings are an aerofoil shape – curved on top and slightly hollow underneath. As the wings move through the air, a difference in air pressure is created above and below, which lifts the bird up into the air. A bird steers by changing the angle of one or both wings, twisting its wings, and spreading and twisting its tail.

Hovering
Hummingbirds are among the birds that hover. They beat their wings in a figure-of-eight pattern, producing lift on both the upstroke and downstroke. They can also fly sideways, straight up and down, backwards, and even upside-down.



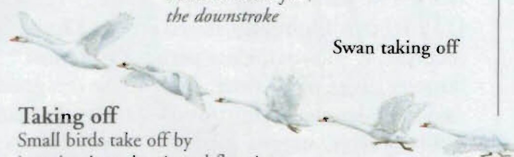


Minla in flight

Forward flight
Most small birds, such as this minla of eastern Asia, fly by flapping their wings up and down. As the wings go down, they push air backwards, moving the bird forwards. As the wings go up, the feathers at the wingtips move apart to allow air to slip through.

Gliding albatross

Gliding
Some large birds rarely flap their wings. Albatrosses and other large seabirds glide on strong winds rising off the waves. Albatrosses can travel for hundreds of kilometres a day. Large land birds, such as vultures and eagles, float on columns of rising hot air called thermals.

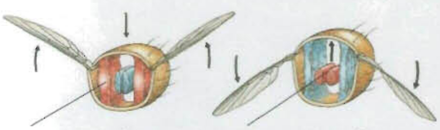


Swan taking off

Taking off
Small birds take off by jumping into the air and flapping their wings. They may take off straight from the ground or from a perch. A large, heavy bird, such as a swan, cannot do this. It needs to run along while flapping its wings to create enough lift for take-off.

Insects

A small insect, such as a mosquito, flaps its wings 1,000 times a second. Most insects flap about 520 times a second. Dragonflies are the fastest insect fliers, reaching nearly 300 km/h (190 mph). Some insects, such as flies, have one pair of wings. Others, such as bees, have two pairs.



Wing muscles
Insect wings developed from their hard body covering. They are not modified legs, like the wings of birds or bats. Insects do not have any muscles on the wings. Instead, their wing muscles are inside the thorax, the middle part of the body.

Cockchafer take-off
The cockchafer is a beetle with two pairs of wings. The front wings are hard wing cases, which are held out of the way during flight. They give the beetle some lift when it flies fast. The flexible back wings flap up and down to provide the power for flight.

Bats

The only mammals able to fly, bats are more acrobatic than birds. They have four large pairs of flight muscles and several smaller pairs, while birds have only two pairs. Each wing consists of skin stretched between four long fingers.

Gliding animals

Some animals can glide slowly downwards. They have developed large fins, or webs or flaps of skin, which they spread out to slow their fall. They have to be able to judge speeds and distances accurately.

Flying squirrel

Flying squirrel
Flaps of skin allow a flying squirrel to glide up to 100 m (330 ft) between trees. The squirrel uses its tail as a rudder, and has sharp claws to grasp the surface on landing.

Flying fish
To escape predators, flying fish swim fast along the surface, then take off and glide for up to 50 m (160 ft), with their huge fins held out.

Flying gecko
The gecko (above) has flaps of skin along the sides of its body and tail. It spreads out the flaps to glide between trees. It has webbed feet to help with steering.

Flying fish

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